

Reflection Properties of the Salisbury Screen

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Abstract—The reflection properties of multiple electric/magnetic Salisbury screens have been studied and analytical formulas developed for the maximally flat design of two and three screen planar stackups, including appropriate values for the spacer materials used to separate the screens. The effect of curvature has also been assessed.

I. INTRODUCTION

THERE IS CONSIDERABLE interest in controlling the reflections of electromagnetic waves from conducting bodies, and quite a few techniques [1]–[3] have been proposed to achieve this goal. One interesting approach is the Salisbury screen [4], which in its most rudimentary form, consists of a lossy screen located a quarter wavelength above a conductor surface. Although this technique was invented many years ago and (as pointed out in the excellent survey in [1]) numerous computer codes exist that can obtain designs numerically, a complete treatment of multiple electric and magnetic Salisbury screens is not available in the archival literature. In this paper, we will, therefore, review the theory and then present analytical formulas for the maximally flat design of multiple electric and magnetic Salisbury screens. Results will be presented for both planar and cylindrical surfaces.

II. REVIEW OF THE SINGLE SCREEN

Consider a thin screen in front of a perfectly conducting plane and separated by a lossless spacer of relative admittance Y_1 as shown in Fig. 1. The screen is assumed to have an arbitrary variation of μ and ϵ in the z -direction, but is homogeneous in the $x - y$ plane. A plane electromagnetic wave is normally incident upon the screen from the left and we desire to find the screen properties and offset d_1 required to produce zero reflection.

Before we can obtain the field reflected by the conducting plane and thin screen, we need to know the boundary conditions that must be applied across the screen. These are derived in Appendix I and are

$$E_x(\Delta) - E_x(0) = -\hat{Z}_m H_y(0), \quad (1)$$

$$H_y(\Delta) - H_y(0) = -\hat{Y}_e E_x(0), \quad (2)$$

where E_x is the tangential electric field, H_y is the tangential magnetic field, $\hat{Z}_m$ is the screen magnetic surface impedance and $\hat{Y}_e$ is its electric surface admittance. These latter quantities

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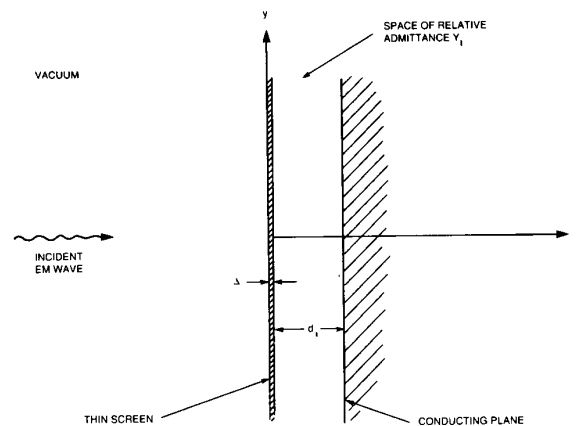


Fig. 1. Single screen in front of a perfectly conducting plane.

are defined as

$$\hat{Z}_m = \omega \int_0^\Delta (\mu'' + j\mu') dz, \quad (3)$$

$$\hat{Y}_e = \omega \int_0^\Delta (\epsilon'' + j\epsilon') dz, \quad (4)$$

where ω is the radian frequency¹ of the incident wave, $\mu \equiv \mu' - j\mu''$ is the complex permeability of the screen and $\epsilon \equiv \epsilon' - j\epsilon''$ is its complex permittivity. Using the boundary conditions in (1) and (2) it is easy to derive the reflection coefficient for an arbitrary number of Salisbury screens in front of a perfectly conducting plane, and these results are summarized in Appendix II. For the case of the single screen shown in Fig. 1 those results give for the field-reflection coefficient²

$$R = R_2 = \frac{j(1 - \beta_1) \sin k_1 d_1 - Y_1(1 - \alpha_1) \cos k_1 d_1}{j(1 + \beta_1) \sin k_1 d_1 + Y_1(1 + \alpha_1) \cos k_1 d_1} e^{j2k_0 d_1} \quad (5)$$

where α_1 and β_1 are the normalized complex magnetic surface impedance and electric surface admittance of the Salisbury screen defined as

$$\alpha_1 = \frac{\hat{Z}_{m1}}{Z_0}, \quad (6)$$

¹ A time variation $\exp(j\omega t)$ is assumed throughout this paper.

² The reflection coefficient is defined as the ratio of the reflected field (in the vacuum region just to the left of the Salisbury screen) to the incident field strength.

$$\beta_1 = \frac{\hat{Y}_{e1}}{Y_0}, \quad (7)$$

with $\hat{Z}_{m1}$ and $\hat{Y}_{e1}$ equal to the magnetic surface impedance and electric surface admittance of the first Salisbury screen, Z_0 is equal to the impedance of vacuum (120π ohms) and $Y_0 = 1/Z_0$. The quantities Y_1 and k_1 are defined as $Y_1 = (\epsilon_1/\mu_1)^{1/2} Y_0^{-1}$ and $k_1 = k_0 (\mu_1 \epsilon_1 / \mu_0 \epsilon_0)^{1/2}$ where ϵ_1 and μ_1 are the permittivity and permeability of the spacer, ϵ_0 and μ_0 are the permittivity and permeability of vacuum and $k_0 = \omega(\mu_0 \epsilon_0)^{1/2} =$ vacuum wavenumber.

We now discuss some special cases of (5).

A. Electric Screen

The classical Salisbury screen is an electric screen located one quarter of an electrical wavelength in front of the conductor. This location is chosen because the electric field is largest there, thus maximizing the electrical dissipation.³ For an electrical screen located a quarter wavelength (at midband) in front of the conducting plane, we set $\alpha_1 = 0$ because the screen has no magnetic properties, and $k_1 d_1 = \pi/2$ because at midband the screen is exactly a quarter of an electrical wavelength from the ground plane. Therefore, at the midband frequency (5) becomes (ignoring the phase term $\exp(j2k_0 d_1)$)

$$R = \frac{1 - \beta_1}{1 + \beta_1}. \quad (8)$$

Consequently, $R = 0$ at midband if $\beta_1 = 1$. That is, $R = 0$ if the electrical conductance of the screen equals the vacuum admittance. However, R is equal to zero only at midband, even if β_1 were frequency independent. In order to explore the behavior of R at other frequencies let us write

$$k_1 d_1 = \frac{\pi}{2} (1 + F) \quad \text{Se aleja um pouco da freq. central} \quad (9)$$

where $F = (f - f_0)/f_0$ and f_0 is the midband frequency. Then for $\beta_1 = 1$ and $F \ll 1$ we get from (5)

$$R \approx \frac{Y_1 \sin \frac{\pi}{2} F}{2j \cos \frac{\pi}{2} F - Y_1 \sin \frac{\pi}{2} F} \approx -j \left(\frac{\pi}{4} Y_1 \right) F. \quad (10)$$

From (10) we see that the magnitude of the reflection coefficient increases linearly as the frequency is moved off mid-band. Note, however, that the slope depends on the relative admittance of the spacer, and decreasing the value of Y_1 increases the bandwidth of the reflection null.

We will see in the next section that it is possible to improve the bandwidth characteristics of R by using two or more screens.

³ Recall that Poynting's theorem is

$$\nabla \cdot \mathbf{S} = -\omega/2 [\mu'' |H|^2 + \epsilon'' |E|^2] - j(\omega/2) [\mu' |H|^2 - \epsilon' |E|^2],$$

where $\mathbf{S} = 1/2 \mathbf{E} \times \mathbf{H}^*$. Therefore the dissipated electric power density is $\omega \epsilon'' |E|^2/2$ and the dissipated magnetic power density is $\omega \mu'' |H|^2/2$.

B. Magnetic Screen

A magnetic screen would be expected to dissipate the most power if it were placed directly on the ground plane, where the magnetic field is a maximum. For this case (5) reduces to

$$R = -\frac{1 - \alpha_1}{1 + \alpha_1}, \quad (11)$$

independent of the value of β (because the electric field $E = 0$ at the conducting plane). Therefore, as noted previously in [1] and [2], there would be zero reflection for all frequencies if we could find a material for which α_1 is unity⁴ and frequency independent. Of course, no such material exists and one must be satisfied with materials which for frequencies in the vicinity of midband have the normalized magnetic surface impedance

$$\alpha_1 \approx 1 + \alpha'(0)F, \quad (12)$$

where $\alpha'(0) = (d\alpha_1/dF)_0$. Now (11) becomes

$$R \approx \frac{\alpha'(0)}{2} F, \quad (13)$$

and we see that $|R|$ increases linearly from zero as the frequency is moved off midband. In a later section we will discuss how a second screen can be used to compensate for the frequency variations in the magnetic surface conductivity.

C. Arbitrary Screen

For a screen in which α_1 and β_1 are not purely real it is still possible to find a location d_1 that gives a zero reflection coefficient. From (5) we see that this occurs if

$$(\beta_1 - 1) = -jY_1(\alpha_1 - 1) \cot(k_1 d_1). \quad (14)$$

For a purely electric screen ($\alpha_1 = 0$) the condition in (14) reduces to

$$\beta_1 = 1 + jY_1 \cot(k_1 d_1). \quad (15)$$

If a spacer with a real permittivity is employed, the result in (15) means that for realizable Salisbury screens (those⁵ with $\text{Im}(\beta_1) > 0$) the spacer thickness will be such that $0 \leq k_1 d_1 \leq \pi/2$, and for $\pi/2 < k_1 d_1 < \pi$ there will be no realizable Salisbury screens that give $R = 0$. Salisbury screens for which $\text{Im}(\beta_1)$ is controlled are called "circuit analog devices," but their design is beyond the scope of this paper.

III. TWO SCREENS

One's ability to produce a broad null in the reflection coefficient is enhanced by using two screens, such as shown in Fig. 2. If we assume that the medium to the left of screen 2 is vacuum (so $Y_3 = 1$), the spacer separations are such that $d_1 = d_2 = d$ and the same material is used in each spacer so that $Y_1 = Y_2 = Y$ and $k_1 = k_2 = k$ then by using the results in Appendix II it is readily shown that the field reflection

⁴ That is, the magnetic surface resistance is equal to 120π ohm.

⁵ That is, $\beta = \hat{Y}_e/Y_0$, so from (4) we see that because ϵ' must be greater than zero then the imaginary part of $\hat{Y}_e$ is also greater than zero.

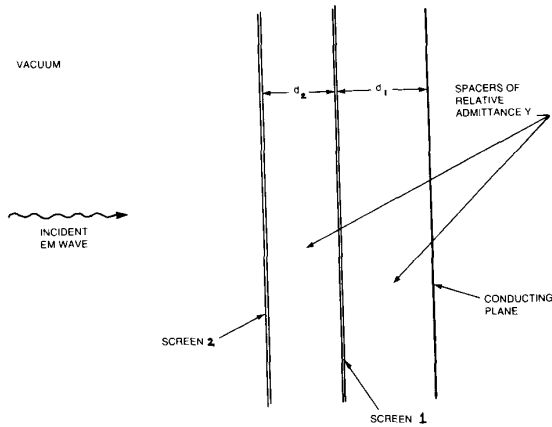


Fig. 2. Two screens in front of a perfectly conducting plane.

coefficient just to the left of screen 2 is

$$R = R_3 = \frac{e^{-j\phi_3}(a_3 + R_2 b_3 e^{j\phi_2})}{c_3 + R_2 g_3 e^{j\phi_2}}, \quad (16)$$

where $\phi_3 = -4k_0d$, $\phi_2 = -4kd$, $a_3 = 1 - Y - \beta_2 + \alpha_2 Y$, $b_3 = 1 + Y - \beta_2 - \alpha_2 Y$, $c_3 = 1 + Y + \beta_2 + \alpha_2 Y$ and $g_3 = 1 - Y + \beta_2 - \alpha_2 Y$. Also R_2 is given by (60) in Appendix II with $\phi_1 = \phi_2 = -2kd$, $a_2 = -\beta_1 + \alpha_1 Y^2$, $b_2 = 2Y - \beta_1 - \alpha_1 Y^2$, $c_2 = 2Y + \beta_1 + \alpha_1 Y^2$ and $g_2 = \beta_1 - \alpha_1 Y^2$. There are a number of situations for which $R = 0$; we will now study some of them.

A. Two Electric Screens

Suppose the two screens have no magnetic properties ($\alpha_1 = \alpha_2 = 0$) and are separated by the same distance ($d_1 = d_2 = d$). We would like to find values for β_1 and β_2 such that at midband both $R = 0$ and $dR/df = 0$ at $f = f_0$ yielding at most a quadratic increase in R from its null value as the frequency is varied from midband. Such a design is sometimes called "maximally flat." If we assume that both β_1 and β_2 are independent of frequency, we find from (16) after considerable algebra that, provided $kd \neq \pi$, both $R = 0$ and $dR/df = 0$ at $f = f_0$ if⁶

$$\beta_2 = 1 - \frac{Y}{2} \left(\frac{3 - \cos \phi_0}{1 - \cos \phi_0} \right)^{1/2} + j \frac{Y \sin \phi_0}{2(1 - \cos \phi_0)}, \quad (17)$$

$$\beta_1 = \frac{2Y[Y \sin \phi_0 - j(\beta_2 - 1) \cos \phi_0]}{-(\beta_2 - 1) \sin \phi_0 + jY \cos \phi_0}, \quad (18)$$

where ϕ_0 is the value of $2kd$ at midband. That is, $\phi_0 = 4\pi f_0 d / (\mu\epsilon)^{1/2}$, where f_0 is the midband frequency and ϵ , μ are the spacer permittivity and permeability. When the screens are separated from each other and the ground plane by a quarter of a wavelength at midband then $\phi_0 = \pi$ and the results in (17)

⁶ The values of β_1 and β_2 were found by expanding R in (9) in Taylor series in F of the form $R = A + BF + \dots$ and then solving for the values of β_1 , β_2 that give $A = 0$, $B = 0$.

and (18) reduce to

$$\beta_2 = 1 - \frac{Y}{\sqrt{2}} \quad (19)$$

$$\beta_1 = \sqrt{2} Y. \quad (20)$$

For $Y = 1$, (19) and (20) reduce to the results cited in [1, sec. 8.6]. Plots of the reflection coefficient versus $F = (f/f_0 - 1)$ for two screens with the values of normalized surface conductance given in (19) and (20) are shown in Fig. 3, with the spacer relative admittance Y as a parameter. Also shown are results for the case when $d_1 \neq d_2$ but rather d_2 is one half-wavelength and d_1 is one quarter wavelength at midband. For this latter case, values of β_1 and β_2 that produce $R = 0$ and $dR/df = 0$ at f_0 are

$$\beta_2 = 1 - \left(\frac{3}{2} \right)^{1/2} Y, \quad (21a)$$

$$\beta_1 = \left(\frac{3}{2} \right)^{1/2} Y. \quad (21b)$$

From Fig. 3 we note two conclusions: 1) two screens separated by a quarter of an electrical wavelength give a broader reflection null than the case when $d_2 = \lambda_{s0}/2$ and $d_1 = \lambda_{s0}/4$ and 2) for the quarter-wavelength screens the bandwidth of the reflection null increases as Y is decreased. Thus, one desires to use a spacer material with as small a value of Y as is feasible (however, $Y < 1$ generally implies that the spacer will be a lossless magnetic material).

B. Inner Screen Electric and Outer Magnetic

We can also consider the case when $d_1 = d_2 =$ one quarter wavelength at midband and the inner screen is electric and the outer magnetic (so that $\alpha_1 = \beta_2 = 0$). In this case one finds from (16) that $R = dR/df = 0$ at $f = f_0$ (or $F = 0$) if

$$\alpha_2 = 1 - \frac{\sqrt{2}}{Y}, \quad (22)$$

$$\beta_1 = \sqrt{2} Y. \quad (23)$$

The resulting reflection coefficient is plotted in Fig. 4 for different values of spacer relative admittance Y . Note that the bandwidth of the reflection null now increases as Y is increased. Also note (by comparing Fig. 4 with Fig. 3) that, unlike the case of two electric screens, the electric/magnetic combination does not necessarily give $|R| = 1$ when $F = \pm 1$.

C. Two Magnetic Screens

Next consider the reflections when a magnetic screen lies directly on the surface of the conducting plane and a second magnetic screen is located an arbitrary distance in front of the ground plane (thus $d_1 = 0$, $d_2 =$ arbitrary). In this case $R = 0$

⁷ λ_{s0} = wavelength in spacer at midband.

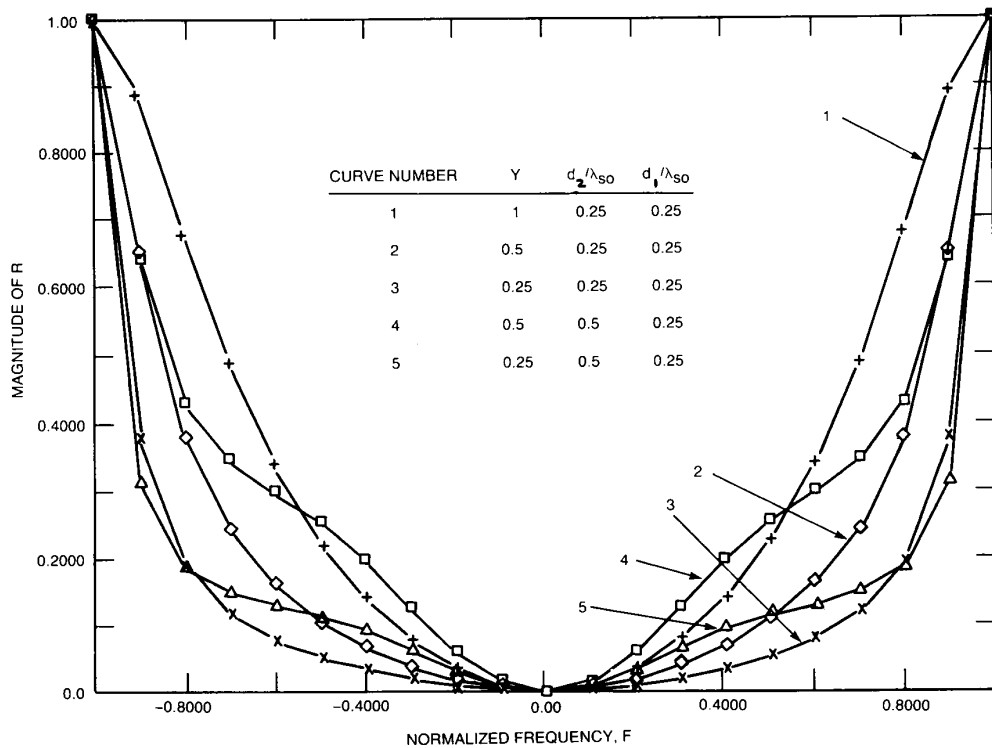


Fig. 3. Magnitude of the normal-incidence reflection coefficient versus normalized frequency for two electric screens in front of a ground plane. Curves are labeled in accordance with table, where λ_{s0} = electrical wavelength in the spacer at midband.

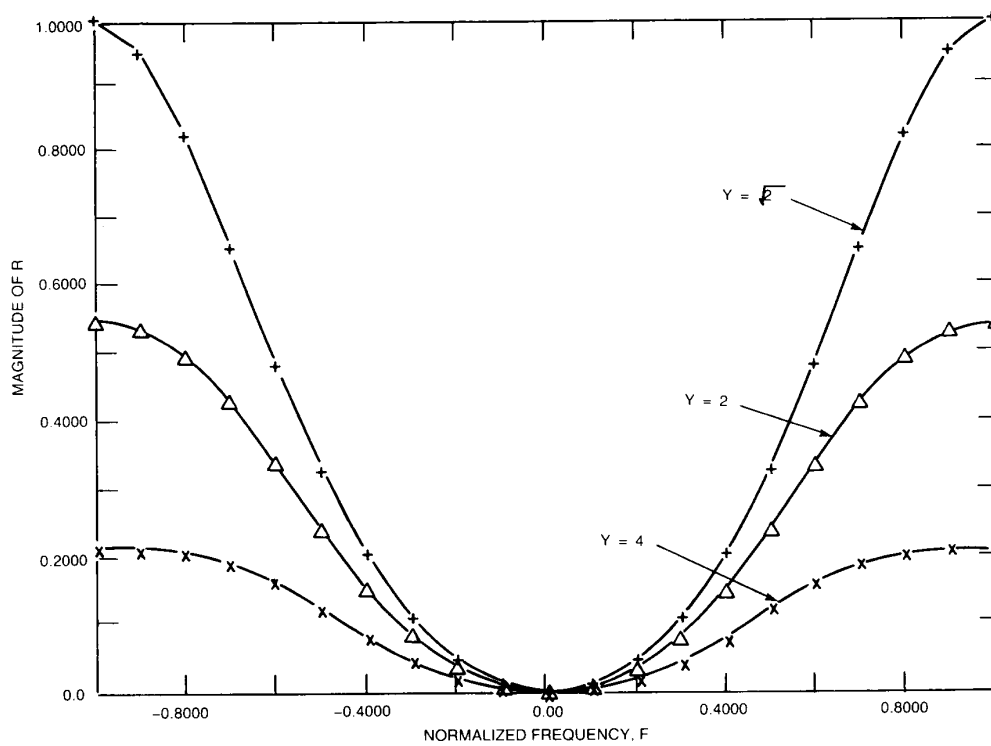


Fig. 4. Magnitude of the normal-incidence reflection coefficient versus normalized frequency for inner electric and outer magnetic screens (with $d_1 = d_2 = \lambda_{s0}/4$, λ_{s0} = midband electrical wavelength in spacer) in front of a perfectly conducting plane. Equations (22) and (23) used for β_1 and α_2 .

for all frequencies if

$$\alpha_2 = 1 - \frac{1}{Y}, \quad (24)$$

$$\alpha_1 = \frac{1}{Y}. \quad (25)$$

The immediate question one might ask is why is the solution independent of the separation of the second screen from the ground plane? This is because the screen on the ground plane absorbs all energy incident upon it if $\alpha_1 = 1/Y$. Therefore, we only require that the outer screen transmit all energy incident upon it, and the condition when this occurs is given by (24).

We now ask "why bother with the outer screen at all if the inner screen absorbs all energy incident upon it?" The answer is that α_1 cannot be a constant for all frequencies (as required for total absorption) so a second screen is needed to compensate for its frequency variations. In order to see how the frequency compensation works suppose α_1 and α_2 are expanded in a Taylor series about their values at $F = 0$, and only the first two terms retained as

$$\alpha_1 \approx \alpha_1(0) + \alpha'_1(0)F, \quad (26a)$$

$$\alpha_2 \approx \alpha_2(0) + \alpha'_2(0)F, \quad (26b)$$

where $\alpha'_1(0) = (d\alpha_1/dF)_0$. If (26a) and (26b) are substituted into (16) and we assume d_2 is equal to one quarter wavelength at midband, we find $R = 0$ and $dR/dF = 0$ at $F = 0$ if both screens are identical, with the properties

$$\alpha_1(0) = \alpha_2(0) = \frac{1}{2}, \quad (27a)$$

$$\alpha'_1(0) = \alpha'_2(0) \equiv \alpha' \quad (27b)$$

and the spacer has the relative admittance

$$Y = 2. \quad (27c)$$

In order to see the difference in reflection produced by use of a second magnetic screen, in Fig. 5 we compare the reflection coefficients of a single magnetic screen and two magnetic screens for the case when the normalized slope $\alpha'/\alpha(0) = 0.5$. It is clear from this figure that the bandwidth of the null is considerably larger for the two-screen arrangement than for the single screen alone.

One could also use an electrical outer screen (instead of the magnetic one discussed above) to provide the compensation for the frequency variations in the magnetic screen on the ground plane. In this case, upon setting $\alpha_2 = 0$ in (16) and letting $\alpha_1 = \alpha_1(0) + \alpha'_1(0)F$, $\beta_2 = \beta_2(0) + \beta'_2(0)F$ we find that R and $dR/dF = 0$ at $F = 0$ if

$$\alpha_1 = \frac{1}{Y}, \quad (28a)$$

$$\beta_2 = 1 - Y, \quad (28b)$$

and

$$\beta'_2(0) = Y^2 \alpha'_1(0) \quad (28c)$$

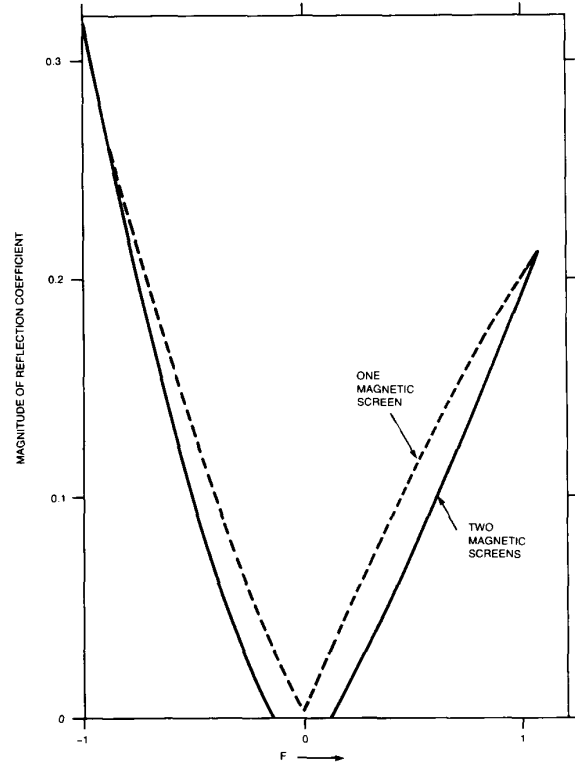


Fig. 5. Magnitude of the normal incidence reflection coefficient versus normalized frequency for one magnetic screen and two magnetic screens (one on conducting plane and the other a quarter-wavelength above it) for the case $\alpha'/\alpha(0) = 0.5$.

if the outer screen is one half-wavelength (at midband) in front of the ground plane and

$$\beta'_2(0) = -Y^2 \alpha'_1(0) \quad (28d)$$

if the outer screen is one quarter-wavelength (at midband) in front of the ground plane. However, it would not appear that using an electric screen for compensation is an easy matter, because it may not be easy to find electric and magnetic materials that fulfill the slope requirements in (28c) and (28d).

IV. THREE ELECTRIC SCREENS

We will now concentrate our attention on purely electric screens, and consider the case of three electric screens equally spaced and separated by identical lossless spacers of relative admittance Y and wavenumber k , as shown in Fig. 6. If we again use the results in Appendix II, it is easy to see that the reflection coefficient in the vacuum region just to the left of the third screen is

$$R = R_4 = \frac{e^{-j\phi_4}(a_4 + R_3 b_4 e^{j\phi_3})}{c_4 + R_3 g_4 e^{j\phi_3}}, \quad (29)$$

where $\phi_4 = -6k_0 d$, $\phi_3 = -6kd$, $a_4 = 1 - Y - \beta_3$, $b_4 = 1 + Y - \beta_3$, $c_4 = 1 + Y + \beta_3$ and $g_4 = 1 - Y + \beta_3$. Also, R_3 is given by (16) except with $a_3 = -\beta_2$, $b_3 = 2Y - \beta_2$, $c_3 = 2Y + \beta_2$ and $g_3 = \beta_2$. R_2 is still given by (60) in Appendix

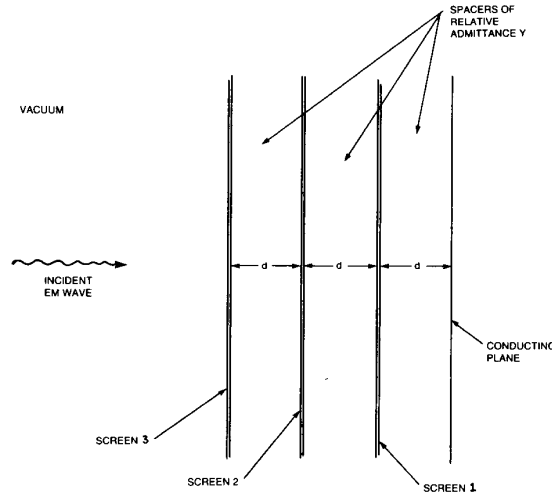


Fig. 6. Three screens in front of a perfectly conducting ground plane.

II except that because all screens are electric we must set $\alpha_1 = 0$ so that $a_2 = -\beta_1$, $b_2 = 2Y - \beta_1$, $c_2 = 2Y + \beta_1$ and $g_2 = \beta_1$.

The maximally flat design involves finding values of β_1 , β_2 , and β_3 that produce $R = dR/df = d^2R/df^2 = 0$ at $f = f_0$ (or $F = 0$). These can be obtained after considerable calculation and satisfy the following three equations:

$$2\beta_1^6 + \beta_1^4(8Y - 19Y^2) + \beta_1^2(8Y^2 - 36Y^3 + 52Y^4) - (12Y^4 - 24Y^5 + 36Y^6) = 0, \quad (32)$$

$$\beta_2 = -\frac{Y^2}{\beta_1} + \frac{Y}{\beta_1} [2(\beta_1^2 - Y^2)]^{1/2}, \quad (33)$$

$$\beta_3 = \frac{\beta_2\beta_1 + Y^2 - \beta_1Y^2}{\beta_2\beta_1 + Y^2}. \quad (34)$$

Thus, (32) must be solved for β_1 and once this is done β_2 and β_3 follow immediately from (33) and (34).

Equation (32) has been solved for β_1^2 using the standard formulas for cubic equations in [5, sec. 28] of Burington, and the results are

$$\text{i) for } Y=1: \beta_3=0.10914, \beta_2=0.51426, \beta_1=1.6441 \quad (35)$$

$$\text{ii) for } Y=0.5: \beta_3=0.52956, \beta_2=0.20158, \beta_1=0.75793. \quad (36)$$

Using the above values of β we have plotted R versus $F = (f - f_0)/f_0$ in Fig. 7. Note, that 1) for $Y = 1$ the bandwidth of the reflection null is larger for the three-screen design than for the two-screen design in Figs. 3 and 2) the bandwidth of the null again increases as the spacer relative admittance Y decreases.

All of the results thus far have been for normal incidence on the screen/ground plane configuration. It is also interesting at this point to study the variation in reflected power as a function

of incidence angle. In Fig. 8 we show the specular reflection coefficient in dB as a function of incidence angle θ (where $\theta = 0^\circ$ for normal incidence) for the above three-screen design with $Y = 1$ and a frequency slightly offset from f_0 . Note that although the reduction in reflection is largest at $\theta = 0^\circ$ there is still a significant reduction for incidence angles as large as 50° or 60° .

V. EFFECT OF CURVATURE

The final topic we wish to consider is the effect of curvature on our results. That is, the results presented thus far are for reductions in reflections from a flat conducting plane, but what happens when Salisbury screens are placed concentrically on a curved surface? In order to explore this point we have studied the reduction in radar cross section when Salisbury screens are placed concentrically on an infinite, perfectly conducting cylinder of radius a . In all the results to follow it has been assumed that the direction of propagation of the incident wave is normal to the cylinder axis, with the incident electric field parallel to the cylinder axis.

In Fig. 9 we show the reduction in the radar cross section when a concentric magnetic screen with normalized magnetic impedance α is placed directly on the surface of the perfectly conducting cylinder. The quantity $\sigma(\alpha)$ is the backscatter radar cross section (per unit length) with the screen present and $\sigma(0)$ is the radar cross section of the bare cylinder. Results are presented for cylinders with $k_0a = 5$ and $k_0a = 30$ where $k_0 = 2\pi/\lambda_0$. Note that the maximum reduction still occurs at $\alpha = 1$ just as for the flat plate, but now there is no true null. Note also that the null depth decreases as the cylinder radius (in wavelengths) decreases. This is as expected.

In Fig. 10 we show equivalent results when a concentric electric screen of normalized electric surface admittance β is placed a quarter of a wavelength above an infinite perfectly conducting cylinder. The spacer relative admittance is $Y = 1$. Note that, for this case, the maximum reduction in backscatter radar cross section no longer occurs at $\beta = 1$ as it does for the planar geometry. That is, for $k_0a = 5$ the maximum reduction

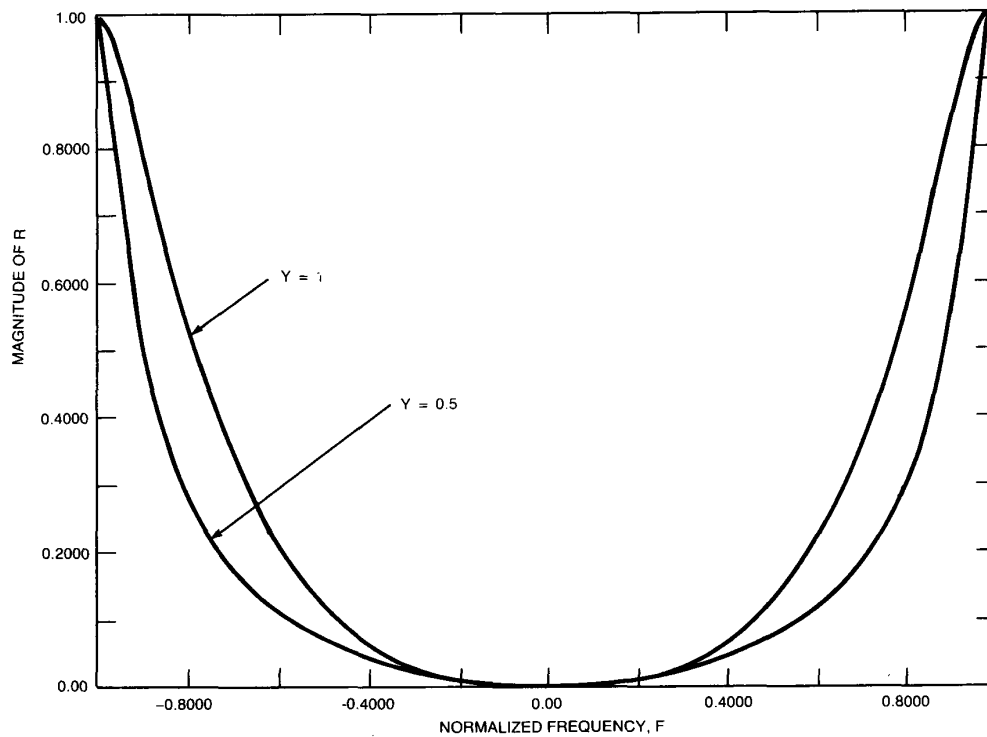


Fig. 7. Magnitude of the normal-incidence reflection coefficient versus normalized frequency for the case of three electric screens (with $\beta_1, \beta_2, \beta_3$ given by (35) and (36)) in front of a perfectly conducting ground plane. $d_1 = d_2 = d_3 = \lambda_{s0}/4$, where λ_{s0} = midband electrical wavelength in the spacer.

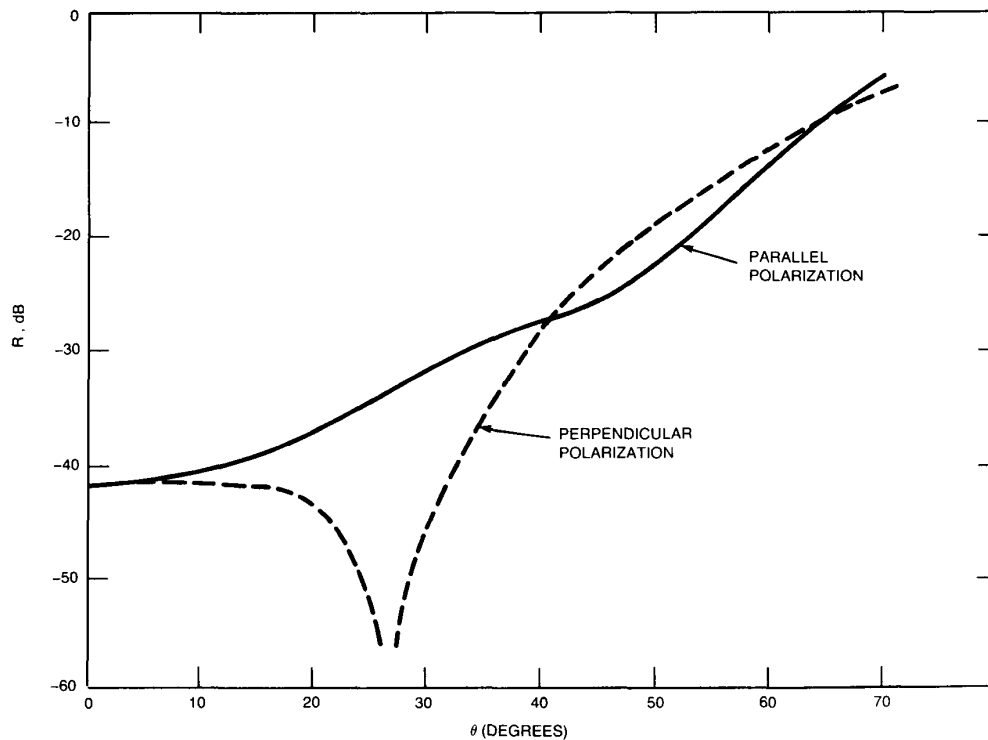


Fig. 8. Magnitude of the specular-reflection coefficient versus incidence angle for the case of three electric screens in front of a perfectly conducting plane. Spacer relative admittance $Y = 1$, $d_1 = d_2 = d_3 = \lambda_{s0}/4$ and $\beta_1, \beta_2, \beta_3$ are given by (35).

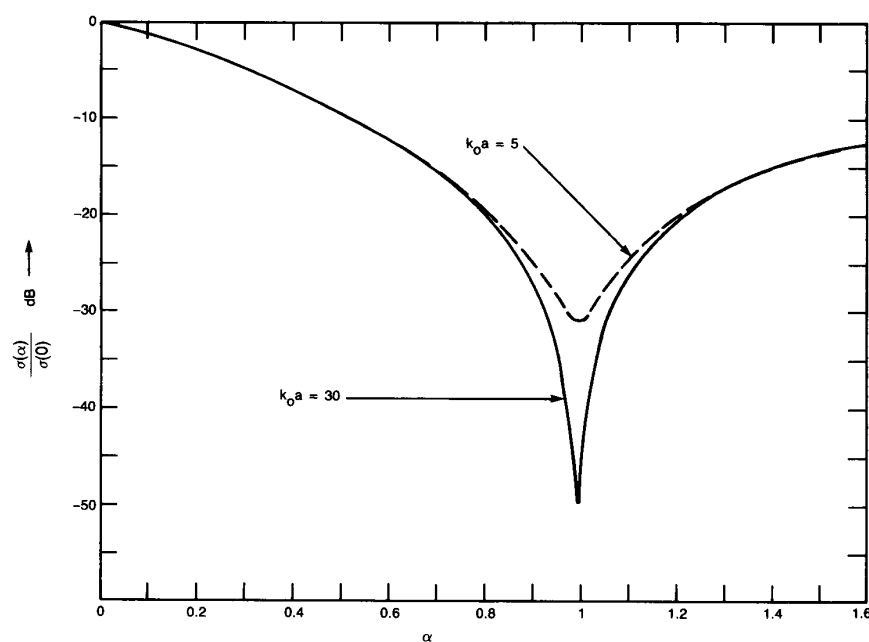


Fig. 9. Reduction in radar cross section of infinite conducting cylinder produced by magnetic Salisbury screen located on cylinder surface (for incident electric field parallel to cylinder axis).

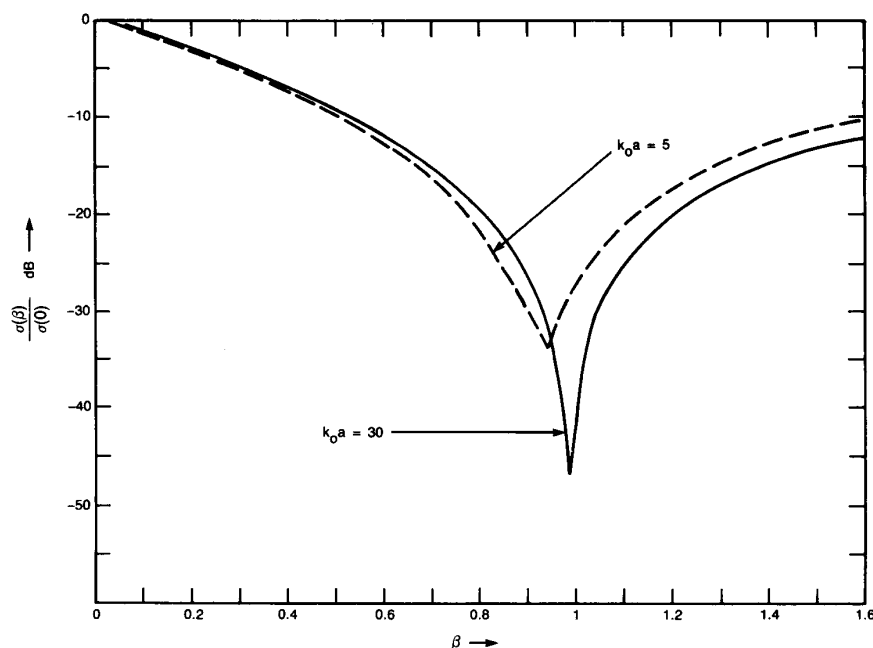


Fig. 10. Reduction in radar cross section of infinite conducting cylinder by concentric electric Salisbury screen located one quarter of wavelength above cylinder (incident E parallel to cylinder axis).

actually occurs at $\beta = 0.94$, but as $k_0 a$ increases the value of β at which the maximum reduction occurs approaches unity, just as was true for the planar geometry.

One final case we can discuss is an infinite perfectly conducting cylinder with an inner concentric electric screen of normalized admittance β_1 located a quarter wavelength above

it and an outer concentric electric screen of normalized admittance β_2 located at a half-wavelength distance from the cylinder surface. The spacer material is assumed to have a relative admittance $Y = 1$. The radar cross section reduction for this case is shown in Figs. 11 and 12. For a planar structure (see (19) and (20)) we would expect a null when $\beta_2 = 1 - 1/$

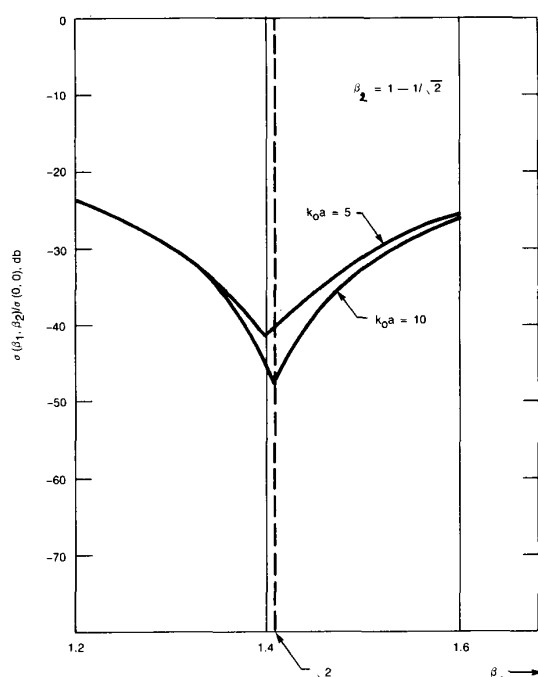


Fig. 11. Reduction in radar cross section of infinite conducting cylinder by two concentric electric Salisbury screens. The inner screen is $\lambda_{s0}/4$ above cylinder and has normalized conductance β_1 , the outer screen is $\lambda_{s0}/2$ above cylinder and has normalized conductance $\beta_2 = 1 - 1/\sqrt{2}$. Spacer relative admittance is $Y = 1$ and incident E is parallel to cylinder axis.

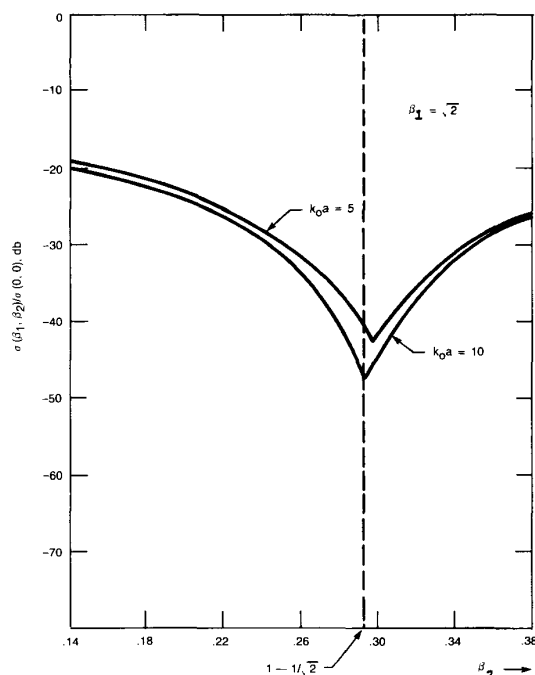


Fig. 12. Reduction in radar cross section of infinite conducting cylinder by concentric electric Salisbury screens. The inner screen is $\lambda_{s0}/4$ above cylinder and has normalized conductance $\beta_1 = \sqrt{2}$. The outer screen is $\lambda_{s0}/2$ above cylinder and has normalized conductance β_2 . Spacer relative admittance $Y = 1$ and incident E is parallel to cylinder axis.

$\sqrt{2}$ and $\beta_1 = \sqrt{2}$. As we see from Figs. 11 and 12 what actually occurs when curvature is included is that, although there is still significant reduction in radar cross section, there is a filling in and shifting of the null as the electrical radius of curvature of the cylinder is decreased. However, as is clear from these and the previous two figures, neither curvature nor ten to twenty percent error in β_1 and β_2 completely destroy the effectiveness of the Salisbury screens in reducing the reflected signal.

VI. DISCUSSION

Although we have explored only a few of the many possible combinations of electric and magnetic Salisbury screens for reducing reflections, we have found some interesting results. We have seen that relatively large reductions in specular reflection are possible over sizeable bandwidths using relatively few Salisbury screens. Furthermore, we have found that these reductions are not completely destroyed if the incident wave is not normal to the surface, if the surface is curved or if we have made errors in fabrication so that the values of α and β are not exactly equal to their design values.

One of the most interesting results of our study is the desirable properties of magnetic materials. In fact, we found that the ideal absorber is a magnetic screen that is placed directly on the conducting surface with the property that $\alpha = 1$, or equivalent (for a spatially homogeneous screen)

$$\omega \mu''(\omega) \Delta = Z_0$$

for all frequencies. Although such materials do not exist, materials can be found that have this property over a relatively narrow frequency band and we found that we could partially compensate for this (thus increasing the bandwidth) by adding a second magnetic screen with an appropriate dielectric spacer.

Because magnetic materials are not always available (or appropriate) we also studied electric Salisbury screens. Even for purely electric screens we found that the bandwidth of the reflection null could be increased if the relative admittance of the spacer were made as small as possible. Because $Y = (\epsilon_r/\mu_r)^{1/2}$ where ϵ_r and μ_r are the relative spacer permittivity and permeability, and ϵ_r is usually greater than unity this implies we would like spacer materials with $\mu_r \gg 1$. In addition, large values of the product $\mu_r \epsilon_r$ are also desirable because the electrical wavelength in the spacer is $\lambda_0(\mu_r \epsilon_r)^{-1/2}$, where λ_0 is the vacuum wavelength. Thus, the geometrical thickness of an N layer Salisbury screen is $0.25 N \lambda_0(\mu_r \epsilon_r)^{-1/2}$ from which we see the larger the value of $\mu_r \epsilon_r$ the thinner the absorber stackup. In general, one desires that the absorber configuration be as thin as possible.

Finally, we note that it really does not take very many Salisbury screens to produce a significant reduction in specular reflection. Thus, for three electric screens with a styrofoam spacer ($Y \approx 1$) we see from Fig. 7 that the reflection coefficient is reduced by a factor of 10 or more over a band $F = -0.5$ to $+0.5$, or equivalently from $f = 0.5$ to $1.5 f_0$, where f_0 is the midband design frequency. Thus if $f_0 = 10$ GHz there would be a tenfold (20 dB) reduction in the field reflection coefficient from $f = 5$ to 15 GHz.

APPENDIX I

DERIVATION OF THE "THIN LAYER" BOUNDARY CONDITIONS

Although the boundary conditions in Eqs. (1) and (2) have been obtained previously [6, 7] by replacing the thin layer by an electric and/or magnetic current sheet we will present an alternate derivation that clearly indicates the conditions under which the thin-screen approximation is valid. Consider a TM wave incident upon a layer at an angle θ , as shown in Fig. 13. The layer is assumed to be homogeneous in the x and y directions but with an arbitrary profile in the z -dimension. If the incident wave has the form $\exp[-jk_0(z \cos \theta + x \sin \theta)]$ then the field in the layer can be written as

$$H_y = h_y(z) e^{-jk_0 x \sin \theta}, \quad (37)$$

$$E_x = e_x(z) e^{-jk_0 x \sin \theta}, \quad (38)$$

$$E_z = e_z(z) e^{-jk_0 x \sin \theta}, \quad (39)$$

where k_0 is the vacuum wavenumber (i.e., $k_0 = \omega/c$, c = speed of light) and h_y , e_x , e_z satisfy

$$\frac{de_x}{dz} = -j\omega\mu\Gamma h_y, \quad (40)$$

$$\frac{dh_y}{dz} = -j\omega\epsilon e_x, \quad (41)$$

$$\omega\epsilon e_z = -k_0 \sin \theta h_y, \quad (42)$$

where

$$\Gamma(z) = \left(1 - \frac{\sin^2 \theta}{\mu_r \epsilon_r}\right), \quad (43)$$

with $\mu \equiv \mu_r \mu_0$, $\epsilon \equiv \epsilon_r \epsilon_0$; ϵ_0 , μ_0 are the permittivity and permeability of vacuum, respectively.

Next, integrate (40) and (41) across the layer. The results are

$$e_x(\Delta) - e_x(0) = -j\omega \int_0^\Delta \Gamma(z) \mu(z) h_y(z) dz, \quad (44)$$

$$h_y(\Delta) - h_y(0) = -j\omega \int_0^\Delta \epsilon(z) e_x(z) dz. \quad (45)$$

We now expand $e_x(z)$ and $h_y(z)$ on the right sides of (44) and (45) in Taylor series about $e_x(0)$ and $h_y(0)$, respectively, using (40) and (41) to calculate $(de_x/dz)_0$, $(dh_y/dz)_0$ and the higher order derivatives. We obtain

$$\begin{aligned} e_x(\Delta) - e_x(0) = & -j\omega h_y(0) \int_0^\Delta \mu \Gamma dz \\ & - \omega^2 \epsilon(0^+) e_x(0) \int_0^\Delta \mu \Gamma z dz \\ & + j \frac{\omega}{2} k_0^2 \mu_r(0^+) \epsilon_r(0^+) \Gamma(0^+) h_y(0) \\ & \cdot \int_0^\Delta \mu z^2 \Gamma dz + \dots, \end{aligned} \quad (46)$$

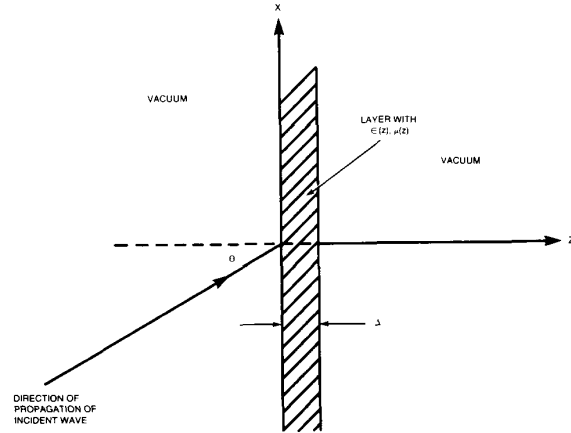


Fig. 13. Geometry used to derive boundary conditions.

$$\begin{aligned} h_y(\Delta) - h_y(0) = & -j\omega e_x(0) \int_0^\Delta \epsilon dz \\ & - \omega^2 \mu(0^+) \Gamma(0^+) h_y(0) \int_0^\Delta z \epsilon dz \\ & + j(\omega/2) k_0^2 \epsilon_r(0^+) \mu_r(0^+) \Gamma(0^+) e_x(0) \\ & \cdot \int_0^\Delta z^2 \epsilon dz + \dots, \end{aligned} \quad (47)$$

where $\epsilon_r(0^+)$ indicates the value of ϵ_r just to the right of the boundary at $z = 0$. We do not need to distinguish $e_x(0^+)$ from $e_x(0)$ because $e_x(0^+) = e_x(0^-)$ and $h_y(0^+) = h_y(0^-)$ by virtue of the continuity of the tangential fields. From (46) and (47) we see that for a TM wave the boundary conditions across the layer can be approximated by

$$e_x(\Delta) - e_x(0) = -j\omega h_y(0) \int_0^\Delta \mu dz, \quad (48)$$

$$h_y(\Delta) - h_y(0) = -j\omega e_x(0) \int_0^\Delta \epsilon dz, \quad (49)$$

provided the following conditions hold

$$1) \sin^2 \theta \ll \mu_r(z) \epsilon_r(z), \quad (50)$$

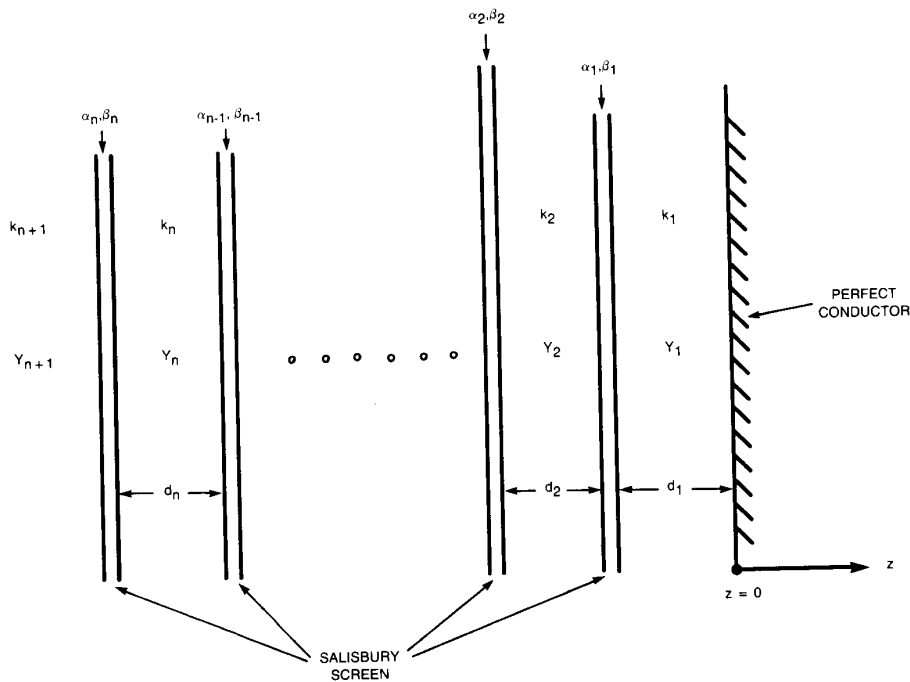
for $0 \leq z \leq \Delta$

$$2) \omega^2 \mu(0^+) \int_0^\Delta z \epsilon(z) dz \ll 1, \quad (51)$$

$$3) \omega^2 \epsilon(0^+) \int_0^\Delta z \mu dz \ll 1, \quad (52)$$

$$4) \frac{\frac{1}{2} k_0^2 \mu_r(0^+) \epsilon_r(0^+) \int_0^\Delta z^2 \epsilon dz}{\int_0^\Delta \epsilon dz} \ll 1, \quad (53)$$

$$5) \frac{\frac{1}{2} k_0^2 \mu_r(0^+) \epsilon_r(0^+) \int_0^\Delta z^2 \mu dz}{\int_0^\Delta \mu dz} \ll 1. \quad (54)$$

Fig. 14. n Salisbury screens in front of a perfectly conducting plane.

Usually, the conditions in (53) and (54) are satisfied if (50) and (51) hold. For a homogeneous layer and θ arbitrary the above conditions reduce to two restrictions

$$\epsilon_r \mu_r \gg 1, \quad (55)$$

$$\frac{k_0^2 \mu_r \epsilon_r \Delta^2}{2} \ll 1. \quad (56)$$

Similar analyses for TE waves leads to the same five restrictions as given in (50)–(54), and the same boundary conditions for the tangential electric and magnetic fields given in (48) and (49).

APPENDIX II

RECURSION RELATION FOR THE REFLECTION COEFFICIENT

Consider a plane wave incident from the left on a stackup consisting of n Salisbury screens separated by n spacers of arbitrary relative admittance as shown in Fig. 14. In the n th layer the electric and magnetic fields are

$$E_n = A_n \exp(-jk_n z) + B_n \exp(jk_n z), \quad (57)$$

$$H_n = Y_n Y_0 A_n \exp(-jk_n z) - Y_n Y_0 B_n \exp(jk_n z), \quad (58)$$

where k_n is the wavenumber in the n th spacer, Y_n is its normalized admittance and $Y_0 = (120 \pi)^{-1}$ is the characteristic admittance of vacuum. If the boundary conditions in (1) and (2) are applied across the n th Salisbury screen and we define the reflection coefficient in the n th spacer by $R_n = B_n/A_n$ it is straightforward to show that the recursion relation between R_{n+1} and R_n is

$$R_{n+1} = \frac{\exp(-j\phi_{n+1})[a_{n+1} + R_n b_{n+1} \exp(j\phi_n)]}{c_{n+1} + R_n g_{n+1} \exp(j\phi_n)}, \quad (59)$$

where

$$a_{n+1} = Y_{n+1} - Y_n - \beta_n + \alpha_n Y_n Y_{n+1},$$

$$b_{n+1} = Y_{n+1} + Y_n - \beta_n - \alpha_n Y_n Y_{n+1},$$

$$c_{n+1} = Y_{n+1} + Y_n + \beta_n + \alpha_n Y_n Y_{n+1},$$

$$g_{n+1} = Y_{n+1} - Y_n + \beta_n - \alpha_n Y_n Y_{n+1},$$

$$\phi_n = 2k_n z,$$

and z is a negative number equal to the distance from $z = 0$ to the left side of the n th Salisbury screen. Because the backing at $z = 0$ is a perfectly conducting plate we have $R_1 = -1$ so the first step in the recursion to calculate R_{n+1} is

$$R_2 = \frac{a_2 - b_2 \exp(j\phi_1)}{c_2 - g_2 \exp(j\phi_1)} \exp(-j\phi_2), \quad (60)$$

where using $z = -d_1$ (remember z is negative because positive z is measured to the right in Fig. 14) we have

$$a_2 = Y_2 - Y_1 - \beta_1 + \alpha_1 Y_1 Y_2,$$

$$b_2 = Y_2 + Y_1 - \beta_1 - \alpha_1 Y_1 Y_2,$$

$$c_2 = Y_2 + Y_1 + \beta_1 + \alpha_1 Y_1 Y_2,$$

$$g_2 = Y_2 - Y_1 + \beta_1 - \alpha_1 Y_1 Y_2,$$

$$\phi_1 = -2k_1 d_1,$$

$$\phi_2 = -2k_2 d_1.$$

In the limit when only a single screen is present and the entire medium to the left of the first screen is vacuum then Y_2

$= 1$ and $k_2 = k_0$ and (60) reduces to the result presented in (5).

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